

Solution Design and Architecture

Reference Data Aggregation Service (RDAS)

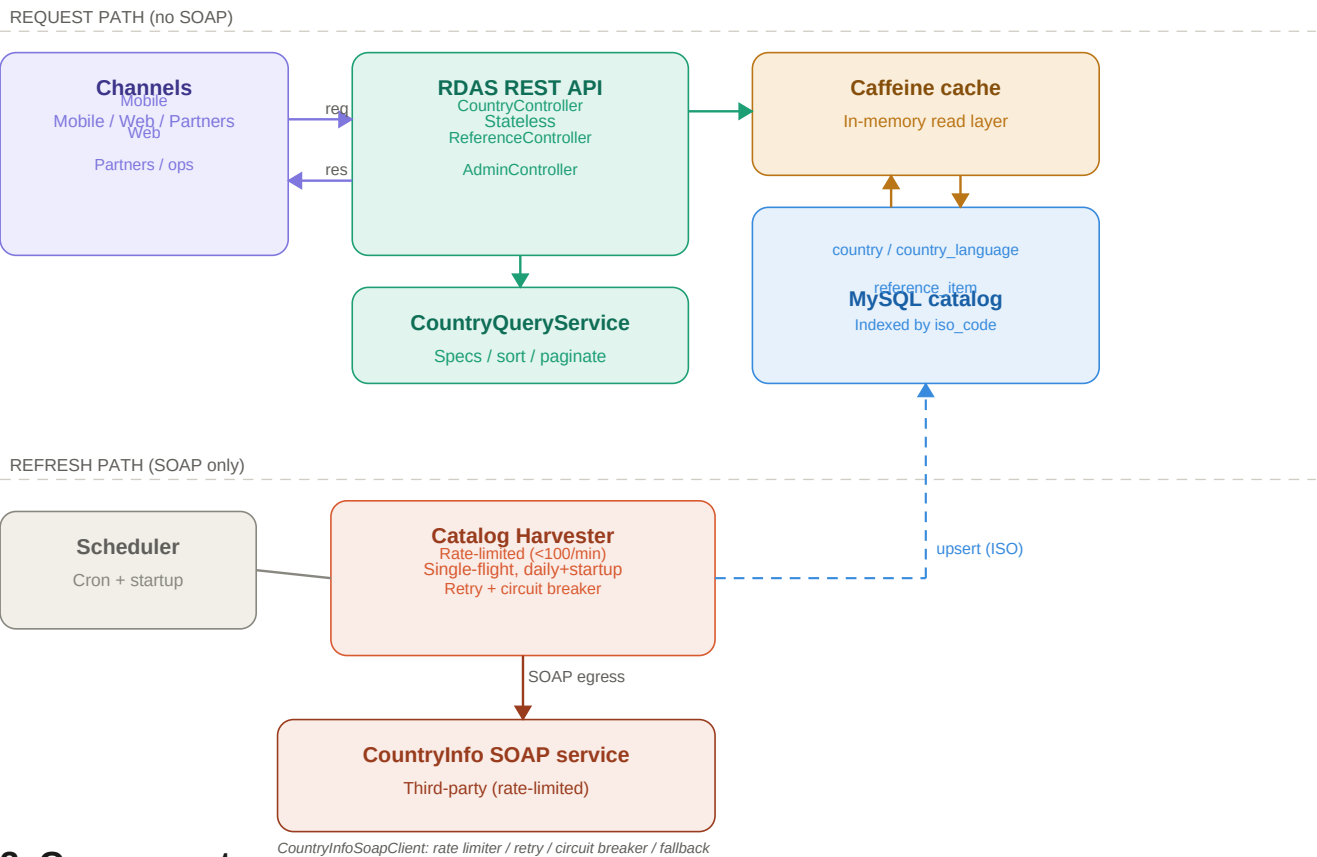
1. Problem

Multiple channels (mobile, web, partner APIs, internal ops) need country, currency, language and geographical reference data. Today each channel calls the third-party CountryInfo SOAP service directly, causing inconsistent responses, poor performance, no filtering/pagination, no auditability, no shared caching, and SOAP credentials spread across many applications. RDAS replaces that with one service that owns the SOAP integration and exposes clean REST/JSON to everyone.

2. Key Architectural Choice: A Materialized Catalog

The SOAP API exposes only narrow, single-record operations and is rate-limited — it cannot search, filter, paginate or sort. RDAS instead materializes a complete local catalog of all countries and reference lists, serving every read from that catalog. SOAP is consumed only by a background harvester.

Architecture Overview



3. Components

Component	Responsibility
CountryInfoSoapClient	The only SOAP caller. Wraps each operation with a rate limiter (<100/min), retry, circuit breaker and fallback.
CatalogHarvestService	Builds/refreshes the catalog: 4 list calls + one FullCountryInfo per country, paced and upserted by ISO code. S
CountryQueryService	Read side. Specification-based filtering, pagination, sorting; Caffeine-cached; maps entities to DTOs. Never ca
Country, ReferenceItem	JPA entities of the catalog (countries + continent/currency/language reference lists).
Controllers	CountryController, ReferenceController, AdminController.
CatalogHealthIndicator	Surfaces catalog size/freshness on /actuator/health (not in the readiness group).

4. Data Model

Entity / Table	Key Fields
Country (table: country)	id, isoCode (unique), name, capitalCity, phoneCode, continentCode, continentName, currencyCode, currencyName, flagUrl, languages [CountryLanguage], createdAt, updatedAt, sourceRefreshedAt
ReferenceItem (table: reference_item)	id, type (CONTINENT CURRENCY LANGUAGE), code, name unique(type, code)
Indexes	iso_code (unique), name, continent_code, currency_code

5. Requirements Mapping

Business Need	Mechanism
Search by country name	name filter, partial case-insensitive LIKE
Filter by continent/currency/language	JPA Specifications (code or name match)
Retrieve country details	GET /countries/{isoCode}
Countries sharing a currency	GET /countries/{isoCode}/currency-peers
Pagination + sorting	Spring Data Pageable, whitelisted sort fields, PagedResponse envelope
Reduce SOAP traffic / caching	Materialized catalog + Caffeine; reads never call SOAP
Auditability	createdAt / updatedAt / sourceRefreshedAt; harvest status and structured logs
One place for SOAP credentials	Only RDAS integrates SOAP; channels never see it

6. Cross-Cutting Concerns

Resilience	Resilience4j (rate limiter, retry, circuit breaker, fallback). Catalog persistence means a SOAP outage does not affect reads.
Observability	Micrometer + Prometheus, Actuator health groups (readiness excludes SOAP), structured JSON logs under the prod profile.
Security boundary	Reference data is public, so ISO codes are exposed directly. Admin endpoints are flagged for auth/network policy in production.
Configuration	12-factor via env vars. H2 for local, MySQL for prod. Same image across environments.
Scalability	Stateless replicas behind a Service and HPA. The harvester is a singleton to keep SOAP load constant regardless of replica count.